

Problem Statement No. 5: Travel Planner Agent

1. The Challenge

A Travel Planner Agent is an AI-powered assistant that helps users plan trips efficiently and intelligently. It uses real-time data to suggest destinations, build itineraries, and recommend transport and accommodation options. By understanding user preferences, budgets, and constraints, it tailors personalized travel plans. Integrated with maps, weather updates, and local guides, it ensures a smooth travel experience. The agent can also manage bookings, alert users to changes, and optimize schedules on the go. This smart assistant transforms complex travel planning into a seamless, enjoyable process.

2. Technology Stack & IBM Cloud Integration

Mandatory Requirement: Use of IBM Cloud Lite Services / IBM Granite is mandatory.

- * AI Model: IBM Granite 3-8B Instruct (ibm/granite-3-8b-instruct) via IBM watsonx.ai REST APIs
- * Cloud Infrastructure: IBM Cloud Lite Account with IAM Authentication & WatsonX Project Instance
- * Backend Framework: Java 17 / Spring Boot 3.2.3 with REST Controller Architecture
- * External Services: Open-Meteo Weather API for real-time forecast data
- * Frontend Interface: Modern Web Application (HTML5, Glassmorphism CSS, JavaScript)

3. Core Agent Capabilities

1. Intelligent Itinerary Generation: Context-aware multi-day travel plans customized by destination and budget.
2. Real-Time Weather Integration: Fetches live temperature, weather conditions, and wind metrics for destination analysis.
3. Curated Destination Search: Instant recommendations for popular tourist spots, budget guides, and luxury travel.
4. Dynamic Live Status: Real-time API health monitor verifying active IBM Granite AI model connection and token auth.

4. Implementation Verification

The application has been fully implemented, tested, and verified on local deployment (http://localhost:8080):

- * IBM Cloud IAM Token Generation: SUCCESSFUL
- * IBM Granite Model Inferencing: LIVE & VERIFIED
- * Weather Data API Integration: LIVE & VERIFIED
- * Responsive Glassmorphic UI: VERIFIED across multiple viewports

5. Working GitHub Repository Link

<https://github.com/Adityapatel-dot/Travel-Planner-Agent-IBM-Cloud>